

American Powerhouses: Economic Dominance within State Lines

GROUP 25

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Goal/Thesis

Our project revolves around the individual states within our nation and their economic performance over time. It's very interesting to us how certain states have really stood above the rest over the past few decades and we want to know why that is. That's why our research will focus on the idea that the economic growth of U.S states are tied to the largest companies, population growth, and R&D spending within their borders. Our hypothesis is that the frequency of industry leaders, specifically Fortune 500 companies in addition to population growth and increases in R&D spending have largely contributed to why certain states top the economic leaderboard in terms of GDP.

Description of Datasets

- Size of the datasets
- Column Names
- Classification of each column
- List of Discrete Values for Categorical Variables
- Distribution of Discrete Variables
- Distribution of Quantitative Variables

1. Annual State GDP and Annual State Population

- The size of the data set is 1428 rows and 6 columns
- The column names are state, year, gdp, population, gdp_1000000, gdp_per_capita
- The classifications are state:categorical, year:discrete, gdp:continuous, population:continuous, gdp_1000000:continuous, gdp_per_capita:continuous
- The discrete categorical values are the years 1997 to 2024.
- The distribution of the quantitative values are:

	year	gdp	population	gdp_1000000	gdp_per_capita
mean	2010.500000	3.415789e+11	6.038048e+06	341578.899580	56565.342648
50%	2010.500000	2.059526e+11	4.213518e+06	205952.650000	52486.258197
std	8.080577	4.310027e+11	6.793717e+06	431002.660054	22033.076736

2. Fortune 500 Company Attributes

- The size of the data set is 1102 rows and 4 columns
- The column names are year, headquarters_state, f500_count, f500_pct_change
- The classifications are year:discrete, headquarter_state: categorical, f500_count: discrete, and f500_pct_change: continuous
- The discrete values for headquarter_state are ['AL', 'AR', 'AZ', 'CA', 'CO', 'CT', 'DC', 'DE', 'FL', 'GA', 'IA', 'ID', 'IL', 'IN', 'KS', 'KY', 'LA', 'MA', 'MD', 'ME', 'MI', 'MN', 'MO', 'MS', 'NC', 'ND', 'NE', 'NH', 'NJ', 'NM', 'NV', 'NY', 'OH', 'OK', 'OR', 'PA', 'RI', 'SC', 'SD', 'TN', 'TX', 'UT', 'VA', 'WA', 'WI']
- The distribution is:

count

headquarters_state AL 28 AR 28 AZ 28 CA 28 CO 28 CT 28 DC 28 DE 28 FL
28 GA 28 IA 28 ID 28 IL 28 IN 28 KS 28 KY 28 LA 28 MA 28 MD 28 MI 28 MN
28 NJ 28 MO 28 NC 28 NE 28 OK 28 OR 28 NY 28 OH 28 WI 28 VA 28 WA 28 TX
28 PA 28 RI 28 TN 28 NV 23 UT 20 SC 20 ME 8 NH 7 SD 6 MS 6 ND 3 NM 1

For the quantitative data:

The units for year is calendar year, f500_count is number of companies, and f500_pct_change is percent change from the previous year

	year	f500_count	f500_pct_change
count	1102.000000	1102.000000	1057.000000
mean	2009.197822	12.649728	0.022205
std	8.070245	14.075210	0.245974
min	1996.000000	1.000000	-0.666667
25%	2002.000000	3.000000	-0.035714
50%	2009.000000	7.000000	0.000000
75%	2016.000000	17.000000	0.039216
max	2023.000000	63.000000	2.000000

3. Annual State R&D Spending

- The size of the data set is 954 rows and 3 columns
- The column names are state, year, research spending

- The classifications are state:categorical, year:discrete, research spending:continuous
- The discrete categorical values are the years 2006 to 2024.
- The distribution of the quantitative values are:

research_spending	
count	9.480000e+02
mean	8.370733e+04
std	3.180530e+05
min	1.500000e+02
25%	8.216196e+03
50%	1.695653e+04
75%	3.392380e+04
max	3.392762e+06

Data Wrangling

- List missing data
- Replacement of missing data (if any)
- Deletion of rows (if any) and reasons
- Check that the data is in the correct range
- Check that the data is in the correct format.
- Check for duplicates and remove them (if any)

1. Annual State GDP

There is no missing data for the GDP dataset. Rows on unnecessary attributes were deleted, like those that were not GDP of the state. All data is in the expected range, the correct format, and has no duplicates.

2. Annual State Population

There is no missing data for the population dataset. No rows were deleted except the total of the United States category. All data is in the expected range, the correct format, and has no duplicates.

3. Fortune 500 Company Attributes

There is missing data for f500_pct_change as the first year of each state is NaN as there is no previous year to compare it to, and so it does not make sense to replace it. We could remove those rows when doing our analysis. All data is in the expected range, the correct format, and has no duplicates.

4. Annual State R&D Spending

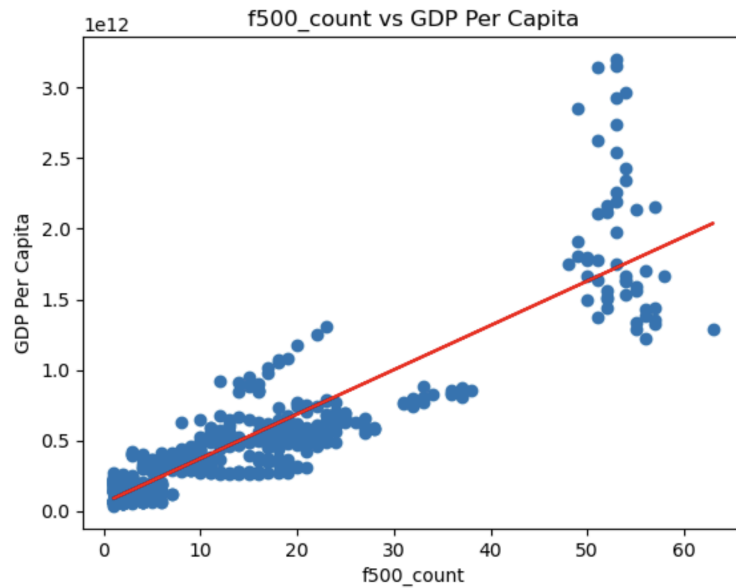
There is no missing data for the state R&D spending. The table was converted from wide to long format. No rows were deleted and all data is in the expected range, the correct format, and has no duplicates.

Preliminary Statistical Analysis (merge all data)

- Determination of the Pearson correlation coefficient among quantitative data
- Determination of the best fit among highly correlated quantities - linear, quadratic, or cubic

	year	gdp	population	gdp_1000000	gdp_per_capita	research_spending	f500_count	f500_pct_change
year	1.000000	0.114740	0.056674	0.114740	0.140137	0.134114	0.028327	-0.009922
gdp	0.114740	1.000000	0.973363	1.000000	0.057890	0.881437	0.894955	-0.032907
population	0.056674	0.973363	1.000000	0.973363	-0.047516	0.813472	0.877896	-0.032240
gdp_1000000	0.114740	1.000000	0.973363	1.000000	0.057890	0.881437	0.894955	-0.032907
gdp_per_capita	0.140137	0.057890	-0.047516	0.057890	1.000000	0.082205	0.059426	-0.020454
research_spending	0.134114	0.881437	0.813472	0.881437	0.082205	1.000000	0.753011	-0.024028
f500_count	0.028327	0.894955	0.877896	0.894955	0.059426	0.753011	1.000000	-0.022050
f500_pct_change	-0.009922	-0.032907	-0.032240	-0.032907	-0.020454	-0.024028	-0.022050	1.000000

The best fit is linear for f500_count and gdp_per_capita



Machine Learning Models

- List and describe the machine learning models you will use to test your hypothesis.

We think that K-Means clustering might be a good fit for our hypothesis. Some states are clearly in a league of their own compared to other states in terms of size and commerce and clustering allows us to look for natural groupings between the states.

We could also use Linear Regression to do a more basic analysis and see the changes of each state over time.

Narration

Storytelling: the next step in visualization

- Data visualizations are normally thought of as tools to analyze data, but Kosara and Mackinlay argue that instead, they should be used to tell memorable stories. Explaining that the way data visualization shines is not by discovering trends but by clearly communicating those trends and findings to people not in the field, such as decision makers.
- In the beginnings of data visualization, most of the research was about the techniques and figuring out which types to use with different data. Which leads us up to modern times,

where we have a great understanding of how to analyse data using visuals, but there is a distinct gap in literature on how to effectively present and communicate those findings effectively.

- One of the ways they argue we should do this is with storytelling, which the author defines as an ordered sequence of steps where each step has some sort of visual or text. Emphasizing that order is extremely important, as it helps create a timeline and help the readers understand the cause and effect of events. For example, most stories are not told linearly, like stories that use flashbacks. But even then, those sub segments are still ordered. So it is important that the order is explained for the story to make sense.
- The author compares this technique to journalism, where they collect a bunch of data from many different sources, but they don't present all of it. They choose the most important parts and guide the reader through the story they want to tell.
- Some examples of how people have used visualization in storytelling throughout time are:
- Florence Nightingale, during the Crimean War, used charts to explain to politicians that the biggest killer was poor sanitation conditions. Her message was able to explain how big of a problem this was, where pure numbers wouldn't have sufficed.
- Another example was John Snow, where he made the infamous cholera map that showed the location of cholera cases in London. Interestingly, he did not make the map to locate the source of the outbreak, as he had already found the water pump responsible with other means, but he made it to act as evidence that the pump was in fact the origin.
- A more modern example was Segel and Heer's study that classified how news media tells stories visually. One of note was the martini glass structure, which is where the narrator structures the data to present their narrative and show some observations, then give you full control over the filters to make your own observations.
- There are also different storytelling scenarios:
- The first is self running presentations to a large audience, like online news visualizations.
- The next is live presentations by a speaker in front of an audience.
- And lastly, there are individual or small group presentations where discussions are possible.

- In conclusion, Kosara and Mackinlay argue that storytelling is the next important step in visualization. Visualization can act as a narrative record of how a finding was found rather than just telling you the result. By using storytelling, analysts can present data in a way that is clear, engaging, and useful for decision-making. As the field becomes bigger, it will be the primary way we pass on knowledge.